

# Skill-suite calibration — mikko- library (2026-06-02) + Spacepotatis + AudiobookMaker audit skills (2026-06-03)

green = saved ≥10% · red = cost ≥10% more · black = within ±10% (no signed signal). Colours the "% saved" columns only; swing/pp and token columns stay neutral.

A/B token measurement (cold arm A vs with-skill arm B) for the 8 cleanly-A/B-able mikko- skills, across **all three models** (Opus 4.8, Sonnet 4.6, Haiku 4.5). This is the **"before" baseline** — the skills were just audited + given freshness blocks (claude-skills #22/#23); a planned optimization pass will re-measure against these numbers. The optimization pass (claude-skills #24) has since landed and three of these skills were re-measured — see **After-optimization re-measure** below. Method mirrors the first calibration document, extended to the full updated suite and every model.

## Method

- **Two arms, same task.** Arm A (cold) scouts from first principles; arm B reads the `SKILL.md`, runs any companion script, follows the procedure. Same deliverable per pair. **48 sub-agent arms** total (8 skills × 2 arms × 3 models).
- **Corpus (8):** `audit`, `ai-codegen-smell-audit`, `react-anti-patterns-audit`, `readme-drift-sync`, `skills-quality`, `skills-freshness`, `skill-usage`, `session-cost`. (Orchestrator/installer/lister/recursive skills excluded as not cleanly measurable.)
- **Tasks auto-synthesized**, pinned to fixed targets: code audits → `mikkonumminen.dev/src/lib` (+ `/terminal`); react → `Spacepotatis`; readme → `mikkonumminen.dev/README.md`; skills-quality/freshness → live `~/ .claude/skills/mikko-*`; skill-usage/session-cost → `~/ .claude/projects`.
- **Single-agent approximation** for orchestrators ( `audit`, `readme-drift-sync` run their parallel passes inline). **Accounting:** per-arm `subagent_tokens` (input+output+cache-creation). Read-only. N = 1 per cell.

## Results — all three models

| Skill                     | Opus A  | Opus B  | Opus % | Sonnet A | Sonnet B | Sonnet % | Haiku A | Haiku B | Haiku % |
|---------------------------|---------|---------|--------|----------|----------|----------|---------|---------|---------|
| skills-freshness          | 101,969 | 23,215  | +77%   | 82,177   | 18,083   | +78%     | 88,412  | 29,948  | +66%    |
| skills-quality            | 75,323  | 21,829  | +71%   | 83,680   | 14,895   | +82%     | 55,753  | 23,637  | +58%    |
| session-cost              | 31,926  | 22,572  | +29%   | 19,823   | 15,859   | +20%     | 69,520  | 23,054  | +67%    |
| react-anti-patterns-audit | 103,254 | 97,460  | +6%    | 73,050   | 68,354   | +6%      | 55,856  | 57,445  | −3%     |
| audit                     | 94,777  | 142,666 | −51%   | 130,314  | 127,089  | +2%      | 64,032  | 59,455  | +7%     |
| skill-usage               | 22,946  | 32,014  | −40%   | 21,899   | 21,938   | ~0%      | 28,879  | 28,147  | +3%     |
| readme-drift-sync         | 33,383  | 44,341  | −33%   | 24,539   | 74,709   | −204%    | 41,126  | 39,715  | +3%     |

| Skill                     | Opus A  | Opus B  | Opus % | Sonnet A | Sonnet B | Sonnet % | Haiku A | Haiku B | Haiku % |
|---------------------------|---------|---------|--------|----------|----------|----------|---------|---------|---------|
| ai-codegen-smell-audit    | 34,224  | 48,665  | -42%   | 26,908   | 52,222   | -94%     | 31,588  | 54,926  | -74%*   |
| Aggregate (ratio-of-sums) | 497,802 | 432,762 | +13%   | 462,390  | 393,149  | +15%     | 435,166 | 316,327 | +27%    |

\* `ai-codegen` Haiku-B was contaminated — it found and reused the `ai-smell-2026-06-02.md` report a *Sonnet* arm wrote earlier (a side-effect-file bleed). Treat that cell as unreliable.

## What the data shows (cross-model)

- Savings scale inversely with model capability** — aggregate +13% (Opus) → +15% (Sonnet) → +27% (Haiku). A weaker model scouts cold less efficiently, so the skill's deterministic procedure / script wins more, relatively. This is the optimization study's and first calibration's core thesis, reproduced cleanly across the whole suite.
- audit is the textbook monotonic case** — -51% → +2% → +7%. On Opus a cold scout is cheap (95K), so the orchestrator-run skill (143K) loses badly; on Haiku the cold scout is dearer relative to the skill, so it flips positive.
- The savings live in the three script-backed skills** — `skills-quality` (+71/+82/+58%), `skills-freshness` (+77/+78/+66%), `session-cost` (+29/+20/+67%) save on every model. They replace LLM reasoning with a Python pre-pass / `scan.mjs`. Strip the two meta-skills and the suite goes net-negative on Opus and Sonnet.
- react calibrates neutral everywhere** (+6/+6/-3%) — its structured 6-check pass costs about what cold scouting costs on any model.
- The big meta-skill saves are "save by not looking."** On all three models the skill arms short-circuited (pre-pass / sha256 hash → "nothing to review") while the **cold arms caught real issues**: bloat (`ai-codegen` 655 lines, `audit`'s 5× duplicated prompt boilerplate), `security-audit` targeting a non-existent codebase, broken cross-repo links, a stale "13 skills" count. The savings are partly a coarser read.
- Haiku is the least reliable arm** — it hallucinated a finding (`skill-calibration` "has no freshness check" — it does), wrongly concluded "token data unavailable" on skill-usage, and one Haiku arm tripped a security flag (ran PowerShell through Bash, circumventing a deny rule). Its verdicts carry the least confidence — same caveat the first calibration flagged.

**Headline:** across all three models the library's token economy is carried by the **script-backed skills** plus the neutral `react`; the prose audits are wash-to-negative against a capable cold model and only turn positive as the model weakens. The cold-arm findings (the bloat list, the real drift) are the targets for the upcoming optimization pass — which should re-measure against these numbers.

## After-optimization re-measure

The "after" the baseline anticipated. *claude-skills* #24 optimized three of these skills — `ai-codegen-smell-audit` (655→580 lines; Provenance extracted to a companion file + 5 dead links removed), `audit` (410→390; the 5×-duplicated sub-agent prompt footer deduped), `readme-drift-sync` (325→310; dated content-calibration narrative extracted). Each optimized skill's **with-skill arm (B)** was re-measured on all three models — same tasks, same accounting. This compares **B(before-optim)** → **B(after-optim)**; a negative  $\Delta$  means cheaper after.

| Skill (arm B)          | Opus →            | Δ    | Sonnet →          | Δ    | Haiku →           | Δ    |
|------------------------|-------------------|------|-------------------|------|-------------------|------|
| ai-codegen-smell-audit | 48,665 → 50,044   | +3%  | 52,222 → 39,148   | −25% | 54,926 → 47,403   | −14% |
| audit                  | 142,666 → 82,841  | −42% | 127,089 → 142,847 | +12% | 59,455 → 56,378   | −5%  |
| readme-drift-sync      | 44,341 → 43,303   | −2%  | 74,709 → 37,044   | −50% | 39,715 → 40,704   | +2%  |
| Aggregate (3 skills)   | 235,672 → 176,188 | −25% | 254,020 → 219,039 | −14% | 154,096 → 144,485 | −6%  |

**Reading: the optimization's per-invocation token effect is below the N = 1 noise floor — this delta is not the optimization.** The per-cell swings span −50% to +12% with no relationship to the 1–3 KB of `SKILL.md` body each skill shed. The same `audit` task cost 56K, 59K, 83K, 127K and 143K across these runs; that ±40–60K task-work variance dwarfs the few-KB body trim by 20–60×. The overall −16% (all nine cells, 643,788 → 539,712) is the *before* pass's high outliers — `audit` Opus 143K, `readme` Sonnet 75K — regressing toward the mean, not a measured saving.

**What this means for skill optimization.** Trimming `SKILL.md` body size buys ≈ 0 measurable per-invocation tokens for task-heavy skills: the one-time body read is a rounding error against the audit work itself. The payoff of the #24 optimizations is real but lives where this metric can't see it — the always-loaded `description` (charged on every matching turn, not just on invocation), context cleanliness, and correctness (the dead links / dead game-refs / stale narrative removed). To *measure* a body-size win you'd need a skill whose body dominates its task (none of these three) or  $N \gg 1$  to drop the noise floor below the signal.

## Spacepotatis audit-skills calibration (2026-06-03)

*A second corpus, same A/B method, run after the Spacepotatis skills were finetuned (quality + freshness, Spacepotatis #286). The 4 cleanly-A/B-able read-only audit skills were measured cold (A) vs with the finetuned skill (B) across all three models — `balance-review`, `content-audit`, `ai-codegen-smell-audit`, `save-roundtrip-audit`. The 8 generative scaffolders (`new-*`, `equipment`) + 2 orchestrators (`modular-architecture-audit`, `security-audit`) were finetuned too but **excluded from A/B** — they mutate the repo, so they aren't cleanly read-only measurable (same exclusion logic the mikko- suite used). 24 sub-agent arms ( $4 \times 2 \times 3$ ). Same accounting ( `subagent_tokens` ),  $N = 1$ , pinned read-only targets. The B arms read the finetuned `SKILL.md` from a worktree off the merged master; the cold A arms are skill-independent and unchanged.*

| Skill (arm B reads the finetuned <code>SKILL.md</code> ) | Opus A  | Opus B  | Opus % | Sonnet A | Sonnet B | Sonnet % | Haiku A | Haiku B | Haiku % |
|----------------------------------------------------------|---------|---------|--------|----------|----------|----------|---------|---------|---------|
| balance-review                                           | 81,911  | 46,403  | +43%   | 60,719   | 25,715   | +58%     | 61,852  | 37,892  | +39%    |
| content-audit                                            | 121,173 | 66,732  | +45%   | 99,250   | 71,408   | +28%     | 62,300  | 57,438  | +8%     |
| ai-codegen-smell-audit                                   | 103,964 | 126,094 | −21%   | 85,301   | 93,056   | −9%      | 61,452  | 62,478  | −2%     |
| save-roundtrip-audit                                     | 64,695  | 72,437  | −12%   | 44,787   | 58,010   | −30%     | 73,877  | 62,038  | +16%    |
| Aggregate (ratio-of-sums)                                | 371,743 | 311,666 | +16%   | 290,057  | 248,189  | +14%     | 259,481 | 219,846 | +15%    |

What this corpus shows:

1. **balance-review** is the **standout** (+43/+58/+39%) — its structured metric procedure (DPS / TTK / energy-per-DPS formulas + a flagged-issues checklist) is far cheaper than scouting the balance maths cold. It's the Spacepotatis analogue of the mikko- script-backed skills.
2. **ai-codegen** is a **wash-to-negative** (-21/-9/-2%) — the same prose-audit pattern as its mikko- copy, reproduced on a different repo. The with-skill arm also follows the skill's "write a report" step, adding output tokens.
3. **content-audit** **saves** (+45/+28/+8%); **save-roundtrip** is **mixed** (-12/-30/+16%) — content's checklist beats cold scouting on every model; save-roundtrip's layer-by-layer procedure costs *more* on Opus/Sonnet (which scout the save pipeline efficiently cold) but saves on Haiku.
4. **Net is a flat** ~+15% (+16/+14/+15%), **not monotonic**. Unlike the mikko- suite's clean +13 → +15 → +27 inverse-capability curve, this corpus blends strong savers (balance, content) with washes (ai-codegen, save), so the per-model spread collapses to a flat band. The durable lesson reproduces across both repos: **token savings concentrate in procedure/script-backed skills; prose audits wash against a capable cold model**.

**Noise-floor demonstration (unintended, but the cleanest in either corpus).** This B column was re-measured. A first pass accidentally read the *pre-finetune* skills (the local checkout lagged the #286 merge), giving an aggregate of +6/+8/+7%; re-running the same arms against the **finetuned** skills — content differing by only a few KB + a freshness block — gave +16/+14/+15%, an ~8-point swing (e.g. content -Opus alone moved 100.9K → 66.7K). Two compounding noise sources explain it, neither a finetune effect: (a) the two skill versions are near-identical — a few KB of body + a freshness block is far too little to move 8 points; and (b) the cold A-arms were *not* re-run, so each cell now pairs a run-1 A against a run-2 B, layering cross-session drift on top of the per-cell **N = 1** variance. The swing is therefore noise, not signal — an accidental, direct confirmation of this doc's own thesis that run-to-run variance dwarfs SKILL.md content. *Which* skill saves vs washes is stable across both runs; only the magnitudes wobble within the noise band.

*Caveats (this corpus):* N = 1 per cell; the with-skill audit arms wrote stray docs/audits/\*-2026-06-03.md reports during measurement (cleaned up); a Haiku content-audit arm tripped the PowerShell-via-Bash deny flag (read-only file listing); the generative + orchestrator skills were finetuned but not A/B-measured. Outcome ≠ tokens — e.g. one Haiku save arm flagged a "critical currentPlanet drop" that the Opus/Sonnet arms (and the skill) correctly read as a by-design re-derivation.

## AudiobookMaker audit-skills calibration (2026-06-03)

A third corpus, same A/B method, on the AudiobookMaker repo (a Python desktop app). The **4 cleanly-A/B-able read-only skills** were measured cold (A) vs with the **finetuned** skill (B) across all three models — *audit*, *ai-codegen-smell-audit*, *release-bundle-audit*, *copyright-scan*. The other 6 skills (*ci-failure-triage*, *pronunciation-corpus-add*, *release-cut*, *voice-pack-finnish*, *work-session*, *worktree-launch*) were finetuned too but **excluded from A/B** — they mutate the repo or need live inputs (a failing CI run, a source recording), so they aren't cleanly read-only measurable (same exclusion logic the other two suites used). **24 sub-agent arms** (4 × 2 × 3). Same accounting ( *subagent\_tokens* ), N = 1, pinned read-only targets — the *src/* tree (audit, ai-codegen), the PyInstaller *.spec* (release-bundle), and a *HEAD~3..HEAD* diff (copyright-scan). The B arms read the finetuned *SKILL.md*; the cold A arms are skill-independent.

| Skill (arm B reads the finetuned SKILL.md) | Opus A  | Opus B  | Opus % | Sonnet A | Sonnet B | Sonnet % | Haiku A | Haiku B | Haiku % |
|--------------------------------------------|---------|---------|--------|----------|----------|----------|---------|---------|---------|
| audit                                      | 115,835 | 100,367 | +13%   | 134,264  | 112,891  | +16%     | 101,262 | 75,794  | +25%    |
| ai-codegen-smell-audit                     | 76,035  | 76,160  | ~0%    | 121,961  | 59,172   | +51%     | 71,418  | 48,614  | +32%    |
| release-bundle-audit                       | 45,668  | 52,984  | -16%   | 73,861   | 65,758   | +11%     | 46,597  | 38,099  | +18%    |
| copyright-scan                             | 25,725  | 30,560  | -19%   | 17,620   | 26,874   | -53%     | 32,365  | 33,988  | -5%     |

| Skill (arm B reads the finetuned SKILL.md) | Opus A  | Opus B  | Opus % | Sonnet A | Sonnet B | Sonnet % | Haiku A | Haiku B | Haiku % |
|--------------------------------------------|---------|---------|--------|----------|----------|----------|---------|---------|---------|
| Aggregate (ratio-of-sums)                  | 263,263 | 260,071 | +1%    | 347,706  | 264,695  | +24%     | 251,642 | 196,495 | +22%    |

What this corpus shows:

- audit** is the **standout** — a **clean monotonic saver** (+13/+16/+25%). Its five-phase robustness procedure caps the cold arm's scaling: the cold arm reads progressively more of the tree as the model weakens, while the skill's bounded sub-agent prompts hold the line. It's the AudiobookMaker analogue of the script-backed savers in the other suites, and the only skill here that saves on every model.
- ai-codegen** **washes on Opus** (~0%) **but saves big on Sonnet/Haiku** (+51/+32%). The cold Sonnet/Haiku arms ran exhaustive audits (122K / 71K tokens); the with-skill arm short-circuited via the skill's prior-audit calibration log ("0 findings, already reviewed") and read far less. Cold Opus was already efficient (76K), so there was nothing to save — the same prose-audit pattern seen in both other repos, here amplified by the calibration short-circuit.
- copyright-scan** is **wash-to-negative** (−19/−53/−5%). Reading the SKILL.md + running its seven-check procedure costs more than a cold scan of a small (2-file) diff — the fixed procedure overhead dominates when the target is tiny. (This skill was previously gitignored as local-only; it was sanitized and brought into the repo for this run.)
- release-bundle-audit** is **mixed** (−16/+11/+18%) — the cold Opus arm traces the spec's reachability efficiently, so the skill's structured Phase 0–2 costs more on Opus, but it saves on the weaker models.
- Net: Opus flat** (+1%), **Sonnet +24%, Haiku +22%; overall +16%**. The inverse-capability curve holds at the aggregate — savings concentrate on the weaker models and in the one procedure-disciplined skill ( **audit** ). The durable lesson reproduces across all three repos: **token savings concentrate in procedure/script-backed skills and scale inversely with model capability; prose audits wash or lose against a capable cold model, especially on a small target.**

*Caveats (this corpus):* N = 1 per cell; **copyright-scan** was local-only (gitignored) and was sanitized + tracked for this run; a pre-existing CLI bug (an engine override inheriting an incompatible configured voice) was fixed in the same finetune branch so the test suite stayed green; the generative + orchestrator skills were finetuned but not A/B-measured. Outcome ≠ tokens — e.g. the **ai-codegen** short-circuit "saves" by trusting a prior calibration log rather than re-deriving findings.

### Caveats

- N = 1 per cell** — direction + rough magnitude; prose-audit magnitudes are noisy (see **readme** : −33/−204/+3%).
- Side-effect contamination** — skill arms wrote reports ( **react-anti-patterns-2026-06-02.md** , **ai-smell-2026-06-02.md** , **SKILL-USAGE-\*.json** ); later same-skill arms occasionally reused them (flagged on **ai-codegen** Haiku-B). These stray files should be cleaned up.
- Auto-synthesized tasks + single-agent orchestrator approximation** inflate **audit** / **readme** skill-arm cost.
- Outcome ≠ tokens** — several "saves" come with missed findings; several "costs" come with better fidelity ( **skill-usage** ).
- Total measurement cost:** ~2.54M **subagent\_tokens** across 48 arms (Opus ~931K, Sonnet ~856K, Haiku ~751K).
- Cross-links: first calibration → **docs/audits/skill-calibration-2026-06-01.md** ; optimization study → **docs/audits/skills-optim-study-2026-05-31.md** .